An orbit is continuous falling

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain a circular orbit with gravity and sideways motion.

SUCCESS CRITERIA - I can...

1. Describe orbit as free fall rather than absent gravity.
2. Distinguish tangent velocity from inward force.
3. Connect circular speed to orbital radius.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A ball launched horizontally at 6 m/s stays airborne 4 s. Find its landing point measured horizontally from the platform base.

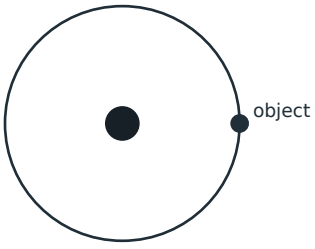
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $GM/r^2 = v^2/r$; $v = \sqrt{(GM/r)}$.
Circular orbit outside a spherical body of mass M; r is orbital radius. Neglect atmosphere and other bodies.

Worked reasoning: A scale-model circular path has radius 4 m and speed 2 m/s. Its period is circumference/speed=$2\pi \times 4/2 = 4\pi$ s. Gravity continuously changes the velocity direction.

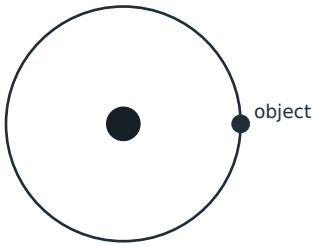
Your turn: Draw a quarter-orbit with tangent velocity arrows and inward gravitational arrows.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw the path of an orbiting object and its velocity/force arrows. Explain how it can fall without immediately hitting the surface.

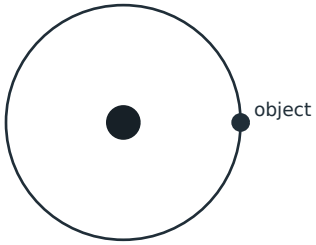


The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

An animation repeats one model orbit every 8 s. How many complete orbits occur in 24 s? Its path length is 40 m per orbit. Find average speed and label the circular model with the supplied information.



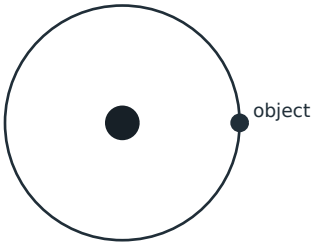
Marzano check - circle: ① ② ③

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05 Explain your reasoning

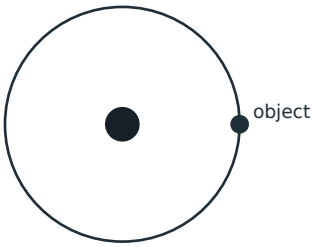
Why does a rocket not need to keep firing forward to remain in an ideal circular orbit?



Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

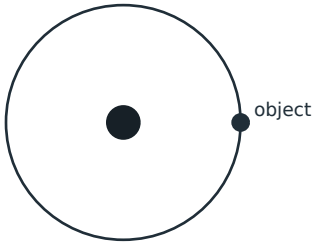
A rover travels around a bend at a steady 6 m/s. Describe an observation that shows velocity changes even though the speed reading does not.
Label velocity and inward acceleration on the circular model.



Marzano check - circle: 1 2 3

07 Exit check

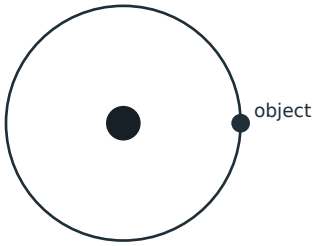
Is an orbiting satellite accelerating if its speed is constant? Draw an arrow to support your answer. Label the radius and show velocity and inward acceleration on the circle.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Compare two model circular paths with the same speed but radii 2 and 4 m. Which takes longer, and by what factor? State that this is a kinematic comparison, not necessarily two orbits around the same body. Label the radius and show velocity and inward acceleration on the circle.



Marzano check - circle: 1 2 3